Simple Simulation II: Psychometrics in AI Deliberation

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1 Experiment Overview

1.1 Aim

The aim of the experiment is simulating **dyadic** civic deliberation on a range of topics, and thereafter judging the deliberative ability of LLMs relative both to their assigned citizen personas, and to the topic itself.

Broadly, this experiment explores the evaluation of deliberative quality using **psychometrics**. Psychometrics is the scientific discipline of measuring psychological attributes of individuals — such as intelligence, personality traits, aptitudes, and attitudes - using mathematical and statistical models.

Generally, psychometrics are measured through administering discrete tests such as the SAT/ACT/Stanford-Binet (for Logical Reasoning) and MMPI/Big-5 (for Personality and Behavior). To operationalize psychometrics within turn-based deliberation, this experiment introduces a post-hoc confidential vote for each agent on a fixed set of policies, with the intention of there being an "ideal policy" (a right answer). While this framework was chosen primarily to challenge popular evaluations of deliberative quality (semantic convergence, DQI, etc.), there is support in general to combine deliberation with voting to optimize democratic outcomes.

Specifically, `Yang and Bachmann 2025` argue that deliberation and voting are complementary rather than conflicting democratic mechanisms. Post-deliberative voting serves as an essential "fair closure mechanism" that guarantees equal standing, encourages authentic preference revelation, and preserves legitimate dissent. Furthermore, `Yang and Bachmann 2025` suggest that administering this ballot privately as a secret vote is essential to prevent social conformity pressure and prevailing-opinion bias.

This experiment defaults to 3 currently relevant issues in Singapore, spanning social, economic, and environmental axes[1].

1.1.1 Topic 1: School Closures

> *Due to declining birth rates, the Ministry of Education (MOE) regularly merges or closes primary and secondary schools. This is a highly emotional topic, pitting practical resource allocation and diverse social mixing against heritage, alumni loyalty, and localized community disruption.*

Reference: CNA, 2021.

1.1.2 Topic 2: Foreign PMET Salary Thresholds

> *This captures the Ministry of Manpower's (MOM) long-running policy calibration between maintaining Singapore's status as a regional corporate/MNC hub and addressing citizen PMET anxieties over wage suppression, displacement, and leadership pipelines.*

Reference: EDB, 2024.

1.1.3 Topic 3: Haze Response

> *Members of Parliament (including WP MP Kenneth Tiong and others) filed parliamentary questions regarding the zero-conviction track record under the THPA since 2014, demanding to know how Singapore can impose real financial accountability on corporate beneficiaries of slash-and-burn clearing without diplomatic paralysis.*

Reference: ST, 2026.

1.2 Setup

1.2.1 LLM Personas

The experiment defaults to 4 different citizen personas:

- $\boldsymbol{P}_1$: The Technocratic Pragmatist; Calm, analytical, focused on feasibility, data, and effecting change through institutional mechanisms.

[1] As will be emphasised later however, the user can configure as many topics as they wish to simulate deliberations for.

- $\boldsymbol{P}_2$: The Ideological Advocate; Articulate, morally committed to preservation and equity; skeptical of technocratic coldness and the status quo.
- $\boldsymbol{P}_3$: The Disaffected Populist; Emotive, candid, feels sidelined by elites; defensive and prone to scapegoating heuristics.
- $\boldsymbol{P}_4$: The Open-Minded Everyman; Plainspoken, humble, constructive; seeks common-sense harmony.

The choice of 4 personas is based on the matrix in Table 1 which crosses *Knowledge* × *Polarization*. **In general however, the experiment allows for any variable number of N personas.**

	Low Polarization	**High Polarization**
High Knowledge	P_1**: Technocrat** Analytical / Feasibility	P_2**: Ideologue** Moral Purism / Principled
Low Knowledge	P_4**: Everyman** Consensus / Pragmatic	P_3**: Populist** Anti-Elite / Emotive

Table 1: 2×2 Persona Matrix.

1.3 Dyadic Configurations

The deliberations are structured around directed pairs of the above citizen personas. In general, for N distinct personas, there exist $\binom{N}{2} = \frac{N(N-1)}{2}$ unique unordered pairings, and $N(N-1)$ possible directed speaking orders (excluding self-dialogue). An example of a dyadic configuration is:

> Technocrat vs Ideologue. This configuration tests the dynamic of "Ideological Friction". The Technocrat optimizes for fiscal viability and mechanism feasibility, while the Ideologue optimizes for moral purity, social justice, or heritage continuity.

1.4 Procedure

1. User prepares a `data/items/` folder with individual `itemXX_{topic}_description.json` files. The user can specify any number of topics, so long as each config is populated with the required information about the topic to be deliberated[2].
2. User prepares a `data/config.json` file and loads in an API key and/or Base URL in `.env`.
3. User executes `python3 -m src.main --config "data/config.json"`.
4. Script programmatically extracts the config details and initializes the inference client for the compute service.
5. For each topic, multi-turn deliberation is conducted between each configured dyadic pair for the specified number of turns.
6. After concluding deliberation, both agents independently cast a secret vote on a 4AFC[3] ballot.
7. Steps 5 and 6 are replicated across T independent trials for the topic-dyad.
8. Raw transcripts for each topic are saved in separate JSON files in `data/expXXX/model_outputs/transcripts/`[4], with format `itemXX_{topic}_transcript.json`[5]. Post-deliberation ballots are saved in `data/expXXX/model_outputs/ballots.json`.
9. Raw psychometric across all topics and trials are saved in `data/expXXX/evals/psychometric_evals.json`. A companion `final_report.md` summarizing outcomes is also generated.
10. Provenance inputs (`config.json` and `data/items/`) are saved in `expXXX/inputs/`, and execution telemetry is saved in `expXXX/metadata/metadata.json`.

1.5 Outcome

In essence, the contract of the repo is:

[2]Note that the use of `items` and "topics" are interchangeable. This will be explained further in Section 1.6.1. This report will use "topics" more.
[3]4-Alternative Fixed Choice: A psychophysical test and measurement method where an observer must choose the correct option out of four alternatives.
[4]The experiment-run indexed `expXXX` folder is dynamically generated.
[5]PDF versions of the transcripts are also programmatically generated for auditing.

> **IF** the user specifies a `data/items/` and `data/config.json` properly and loads in a valid API key and/or Base URL in `.env`,
>
> **THEN** the pipeline will run an N-turn deliberation between M user-specified dyads of agent personas over W topics, replicated for T-trials. Each trial will end with a final policy vote, and then the chat histories and voting outcomes of the trials are recorded in a dedicated folder. The pipeline will also output a `psychometric_evals.json` with a series of psychometric figures and a `final_report.md` with overall conclusions. All experiment outputs will be saved in an automatically created & indexed experiment run sub-directory of `data`.

In general, if the user requests W topics of deliberation between M-configurations of dyads to be replicated for T-trials per topic, then there will be a total of $W \times M \times T$ deliberations generated, and $W \times M \times T \times 2$ ballots conducted (since each of the 2 agents in each deliberation vote independently).

A sample `itemXX_{topic}_transcript.json` looks like:

```
[
  {
    "item_id": "ITEM_01",
    "item_title": "School Closures and Mergers in Singapore",
    "dyad_id": "D1",
    "agents": [
      "technocrat",
      "ideologue"
    ],
    "trial_index": 0,
    "turns": [
      {
        "turn_index": 0,
        "speaker": "Technocratic Pragmatist (P1)",
        "persona_id": "technocrat",
        "statement": "..."
      },
      {
        "turn_index": 1,
        "speaker": "Ideological Advocate (P2)",
        "persona_id": "ideologue",
        "statement": "..."
      },
      ...
    ]
  },
  {
  ...
  }
]
```

A sample `ballots.json` looks like:

```
[
  {
    "item_id": "ITEM_01",
    "dyad_id": "D1",
    "trial_index": 0,
    "votes": {
      "technocrat": {
        "item_id": "ITEM_01",
        "dyad_id": "D1",
        "trial_index": 0,
        "persona_id": "technocrat",
        "raw_response": "C",
        "displayed_letter": "C",
```

```
        "selected_key": "A",
        "is_correct": true,
        "is_valid": true,
        "display_mapping": {
          "A": "C",
          "B": "D",
          "C": "A",
          "D": "B"
        }
      },
    }
  },
  {
  ...
  }
]
```

A sample `data/expXXX/evals/psychometric_evals.json` looks like:

```
{
  "ctt": {
    "ITEM_01": {
      "item_id": "ITEM_01",
      "difficulty_p": 1.0,
      "discrimination_index": 0.0,
      "distractor_counts": {
        "A": 48,
        "B": 0,
        "C": 0,
        "D": 0
      },
      "distractor_ratios": {
        "B": 0.0,
        "C": 0.0,
        "D": 0.0
      }
    },
    ...
  },
  "irt": {
    "items": {
      "ITEM_01": {
        "item_id": "ITEM_01",
        "difficulty_b": -1.6264,
        "discrimination_a": 2.2961
      },
      ...
    },
    "deliberators": {
      "dyad_theta": {
        "D1": 0.2033,
        "D2": 0.2033,
        "D3": 0.2033,
        "D4": 0.2033,
        "D5": 0.2033,
        "D6": 0.2033,
        "D7": 0.2033,
        "D8": 0.2033
      },
      "individual_theta": {
        "everyman": 0.1155,
```

```
        "ideologue": 0.1155,
        "populist": 0.1155,
        "technocrat": 0.1155
      },
      "persona_by_dyad_theta": {
        "everyman": {
          "D4": 0.1155,
          "D5": 0.1155,
          "D7": 0.1155,
          "D8": 0.1155
        },
        ...
      }
    }
  },
  "individual_success_rates": {
    "technocrat": 1.0,
    "ideologue": 1.0,
    "populist": 1.0,
    "everyman": 1.0
  },
  "insights": {
    "hardest_item": "ITEM_01",
    "easiest_item": "ITEM_01",
    "most_discriminating_item": "ITEM_01",
    "lowest_competence_dyad": "D1",
    "highest_competence_dyad": "D1",
    "average_dcr_by_distractor": {
      "B": 0.0,
      "C": 0.0,
      "D": 0.0
    },
    "most_tempting_distractor": "B",
    "total_votes_cast": 144,
    "valid_votes_count": 144,
    "discarded_ballot_count": 0
  }
}
```


A sample `data/expXXX/metadata/metadata.json` looks like:


```
{
  "experiment_name": "Psychometric Deliberative Capacity Benchmark",
  "model": "deepseek-ai/DeepSeek-V4.1-Flash-1017",
  "cost_usd": 0.1118,
  "runtime_seconds": 699.0,
  "runtime_formatted": "11m 39.0s",
  "prompt_tokens": 151200,
  "completion_tokens": 54930,
  "total_tokens": 206130,
  "timestamp": "2026-10-30T21:13:05"
}
```


1.6 Evaluation

1.6.1 Variables

Independent variables:

- The policy dilemma topics.
- The citizen personas.

- The dyadic configurations of personas.
- The LLM of choice.

Dependent variables: The dependent variables are **psychometric** variables, influenced by Classical Test Theory (CTT) and Item Response Theory (IRT). In this context, the `item` is the topic of deliberation and the following metrics are evaluated based on the secret ballots that the agents place after deliberation.

CTT Metrics:

1. **Consensus Rate**: The proportion of the deliberations where both participants in the dyad reach consensus by independently voting for the ideal policy choice on their secret ballots.
2. **CTT Item Discrimination Index**: Measures a topic's ability to differentiate between high-competence and low-competence dyads by taking the difference of the Consensus Rate of the top and bottom 27% of the cohort.
3. **Distractor-to-Correct Ratio**: Quantifies the relative pull of a bad-faith option against the ideal choice.

IRT Metrics:

1. **Item Difficulty (b_i)**: The latent deliberative ability required for a deliberative unit (dyad or individual agent) to have a 50% probability of voting for the ideal choice in the secret ballot.
2. **IRT Item Discrimination (a_i)**: The slope of the Item Characteristic Curve (ICC) at the Item Difficulty, reflecting how sharply the topic differentiates between differing ability levels.
3. **Latent Deliberative Ability (θ)**: The standardized competence trait of a deliberative unit, estimated via regularized maximum a posteriori (MAP) estimation under a zero-mean Gaussian prior ($\theta \sim \mathcal{N}(0, \sigma^2 = 2.0)$):
   - **Dyadic Deliberative Ability (θ_{dyad})**: Measures joint consensus convergence where both agents in the dyad simultaneously vote for the ideal choice. The reported dyadic ability θ_{dyad} is the arithmetic mean across all T replication trials:
   $$\theta_{\text{dyad}} = \frac{1}{T} \sum_{t=0}^{T-1} \theta_{(d,t)}$$
   Under 100% consensus saturation (where every achieves consensus, $Y_{j,i} = 1$ everywhere), the likelihood is monotonic without an interior maximum; the parameters b_i, a_i, θ are structurally not identified by empirical variance and settle at the regularizing prior boundary mode.
   - **Individual Deliberative Ability (θ_{persona})**: Measures a persona's propensity to vote for the ideal choice across all topics. The reported individual ability θ_{persona} is the arithmetic mean across all individual deliberator sessions. This metric is also capable of saturating under 100% ideal-choice selection rate for an individual agent.
   - **Partner Interaction Effects ($\theta_{\text{persona}|\text{dyad}}$)**: Conditions an individual persona's latent ability on the specific dyadic partner, isolating communicative synergy and polarization shifts.

The reason we judge deliberative ability on three levels:

- Individual ability measures the deliberative quality of a specific persona group relative to the topic. It represents the ability of an individual to identify a "reasonable"option after deliberation.
- Partner Interaction Effects recognizes the possibility that an individual's deliberative quality is affected by the behavior of those they deliberate with, and thus nuances the Individual Deliberative Ability.
- Dyadic ability measures the deliberative quality of a specific configuration of 2 personas, relative to topic. It is similar to the Partner Interaction Effect except that it aggregates latent ability on the basis of dyadic convergence towards the "ideal"option, instead of examining each individual persona's total number of "ideal"votes conditioned on different past dyadic partners.

It should be clear now, as has been prefaced in Section 1.4, why "topics" and `items` are interchangeable. To repeat, the "Item" in IRT is the "topic"(the difficulty of which we're testing) and the "Response" in IRT is the ballot vote (through which we judge the latent deliberative ability of an agent or dyad).

1.6.2 Interpretation

1. Consensus Rate (p_i): Measured between [0.0, 1.0]; the larger this metric, the easier the topic is for achieving consensus.
2. CTT Item Discrimination: Measured between [-1.0, +1.0]; the larger this metric, the greater the ability of the topic to discriminate deliberative agents.
3. DCR Ratio: Measured between [0.0, $+\infty$]; under 4AFC random guessing, expected DCR = 1.0; values above 1.0 indicate distractors that out-compete the integrative compromise.
4. Item Difficulty (b_i): Measured between [-4.0, +4.0] under regularized Gaussian prior; the larger this metric, the greater the latent ability required to tackle this topic.
5. IRT Item Discrimination (a_i): Measured between [0.1, 4.0]; the larger this metric, the greater the slope and ability of the topic to discriminate deliberative agents by ability.
6. Latent Ability (θ): Measured between [-4.0, +4.0] under regularized Gaussian prior; the larger this metric, the greater the agent's latent ability (relative to the class of ability measured - ref above).

1.6.3 Theoretical/Empirical Basis

The application of CTT/IRT theory to LLM-simulated deliberation was motivated by `Ke et al. 2026`, which used it to evaluate LLM responses to assessments meant for students, to judge the quality of such assessments.

`Bächtiger et al. 2022` also suggest that the very popular quantitative metric, DQI, which has been used to judge LLM-deliberative quality in `Zhuang et al. 2026`, may also be reformed using IRT; making deliberative quality a latent instead of absolute variable.

To ground the metrics themselves, CTT Item Discrimination is explained in greater detail in `Kelley 1939`, DCR in `Haladyna 2004`, and IRT parameters in `Baker 2001`.

2 Pilot Experiments Specifics

This section describes the specific conditions and configurations under which four pilot experiments were run. The experiments are not meant to produce statistically or theoretically very robust results, but represent a proof-of-concept for incorporating IRT/CTT and post-deliberation voting in LLM-simulated deliberations.

The four pilot experiments are almost entirely identical in configuration, differing only in the underlying LLM used for the deliberative agents. This will be further discussed in Section 2.3.

2.1 Repo Architecture

```
simple-simulation-psychometric-eval-of-deliberation/
├── data/
│   ├── baselines/ ..... Pre-deliberation counterfactual baseline ballots across all 4 models
│   │   ├── baseline_deepseek_ai_deepseek_v4.1_flash_1017.json
│   │   ├── baseline_openai_gpt_oss_120b.json
│   │   ├── baseline_qwen_qwen3.5_9b.json
│   │   └── baseline_meta_models_muse_spark_open_816b_a42b.json
│   ├── config.json ..... Master simulation configuration (models, dyads, items, IRT priors)
│   ├── items/ ..... Standardized policy dilemma items
│   │   ├── item01_school_mergers_description.json ..... School Mergers policy item
│   │   ├── item02_foreign_talent_description.json ..... Foreign PMET Thresholds item
│   │   └── item03_transboundary_haze_description.json ..... Transboundary Haze Enforcement item
│   ├── exp000/ ..... Experiment 0: deepseek-ai/DeepSeek-V4.1-Flash-1017 (552B MoE)
│   │   ├── inputs/ ..... Immutable snapshot of inputs for run reproducibility
│   │   │   ├── config.json ..... Snapshot of master configuration used for run
│   │   │   └── items/ ..... Snapshot of standardized battery items
│   │   ├── model_outputs/ ..... Raw model responses and ballots
│   │   │   ├── ballots.json ..... Post-deliberation secret votes and parsed selections
│   │   │   └── transcripts/ ..... Item-by-item multi-turn dialogues (.json and ReportLab .pdf)
│   │   ├── evals/ ..... Psychometric calibration and comparative reporting
│   │   │   ├── psychometric_evals.json ..... Computed CTT ($p, D$, DCR) and 2PL IRT ($\theta, b, a$) metrics
│   │   │   └── final_report.md ..... Synthesized markdown report of benchmark results with Mermaid diagrams
│   │   └── metadata/ ..... Execution operational telemetry
│   │       └── metadata.json ..... Cost ($0.1118), wall-clock runtime (11m 39s), and token consumption
│   ├── exp001/ ..... Experiment 1: openai/gpt-oss-120b (117B MoE, 5.1B active)
│   │   ├── inputs/ ..... Configuration snapshot and items battery
│   │   ├── model_outputs/ ..... Ballots and multi-turn transcripts (.json and .pdf)
│   │   ├── evals/ ..... Psychometric calibration and markdown report
│   │   └── metadata/ ..... Cost ($0.0525), wall-clock runtime (13m 45s), and token metrics
│   ├── exp002/ ..... Experiment 2: Qwen/Qwen3.5-9B (9B Dense, Reasoning Disabled)
│   │   ├── inputs/ ..... Configuration snapshot with enable_reasoning: false
│   │   ├── model_outputs/ ..... 100% valid ballots and unmediated dialogue transcripts
│   │   ├── evals/ ..... Psychometric calibration and consensus verification
│   │   └── metadata/ ..... Cost ($0.0421), wall-clock runtime (9m 14s), and token metrics
│   └── exp003/ ..... Experiment 3: meta-models/Muse-Spark-Open-816B-A42B (816B MoE)
│       ├── inputs/ ..... Configuration snapshot and items battery
│       ├── model_outputs/ ..... Ballots and multi-turn transcripts (.json and .pdf)
│       ├── evals/ ..... Psychometric calibration and markdown report
│       └── metadata/ ..... Cost ($1.0055), wall-clock runtime (34m 06s), and token metrics
├── docs/
│   ├── coding_guidelines.md ..... Architectural rules, typing standards, and testing protocols
│   ├── experiment_details.tex ..... Comprehensive formal experimental documentation in LaTeX
│   ├── experiment_design_psychometric_eval_deliberation.md .... Psychometric benchmark evaluation framework
│   ├── prompt_templates.md ..... Citizen persona system prompts and ballot templates
│   └── reviewer_feedback_response.md ..... Systematic rebuttal addressing theoretical and methodology critiques
└── src/
    └── main.py ..... Master orchestrator (execute_simulation_run) and CLI entry point
```

- deliberation.py ..... Multi-turn dyadic deliberation chat loop engine
- ballot.py ..... Secret ballot administrator with option randomization and robust parser
- baselines.py ..... Pre-deliberation counterfactual baseline voting orchestrator
- psychometrics.py ..... Classical Test Theory and 2PL IRT regularized L-BFGS-B estimation engine
- personas.py ..... Citizen persona system prompts, traits, and metadata resolution
- report.py ..... Synthesis markdown and Mermaid workflow diagram generator
- transcripts_pdf.py ..... ReportLab PDF transcript document generation engine
- schemas.py ..... TypedDict structural definitions enforcing static type safety
- llm.py ..... OpenAI-compatible client with environment credential validation

tests/

- test_ballot_parser.py ..... Unit tests for ballot option extraction and presentation randomization
- test_deliberation.py ..... Unit tests for dialogue turn structure and history assembly
- test_model_availability.py ..... Integration test verifying configured LLM availability on endpoint
- test_pipeline.py ..... Full end-to-end dry-run pipeline and CLI execution tests
- test_psychometrics.py ..... Unit tests for CTT formulas, 2PL IRT monotonicity, and prior bounds
- test_transcripts_pdf.py ..... Unit tests for PDF transcript rendering and style flowables

- pyproject.toml ..... Pytest, mypy, and build configurations
- requirements.txt ..... Pinned runtime dependencies (numpy, scipy, openai, pytest, reportlab)
- README.md ..... Repository documentation, psychometric guide, and execution tutorial

2.2 Overall User Configurability

`itemXX_{topic}_description.json`:

- `item_id`
- `title`
- `topic_prompt` (user prompt)
- `ballot`: `options`, `correct_key`

`config.json`:

- `experiment_metadata`: name, description, version
- `model_config`: model assignment (mapping the deliberative agent model to the desired LLM)[6] and model settings (`enable_reasoning, reasoning_effort, temperature, deliberation_max_tokens, ballot_max_tokens`)[7].
- `system_prompts`: agent persona prompts
- `user_prompts`: deliberation turn signal, secret ballot invitation.
- `deliberation_config`: turns per deliberation, trials per topic, seed for mcq option order randomization, dyad configurations.

2.3 Model Config

The models chosen for the 4 experiments were:

- `deepseek-ai/DeepSeek-V4.1-Flash-1017` (552B MoE, 37B active)
- `openai/gpt-oss-120b` (117B MoE, 5.1B active)
- `Qwen/Qwen3.5-9B` (9B dense)
- `meta-models/Muse-Spark-Open-816B-A42B` (816B MoE, 42B active)

The motivation for testing very differently sized models is grounded by `Ke et al. 2026` who suggest that larger models are likelier to ignore specific persona prompting and default to pretrained knowledge levels and behavioral propensities, hence complicating the ability to realistically simulate assessment (or in this context, deliberation) difficulty. Similarly, `Tan et al. 2025` caution that larger models are often resistant to harboring counterfactuals or misleading information (due to extensive

[6] Example, `agent_model`: `openai/gpt-oss-120b`

[7] The validity of these settings, especially in terms of verbatim field names for `reasoning_effort` and `temperature`, is of course dependent on the service being used to run inference on LLMs. This experiment uses `together.ai`

safety alignment), while smaller models are instead too vulnerable to authoritative corrections (whether factual or not).

In a different vein, Park et al. 2023 have cautioned that very large/frontier models are often susceptible to their extremely extensive RLHF training, which primes them for conversational agreeableness (as chatbots) while Veri 2026 goes further to suggest that such models converge quickly to certain sociopolitical centers (usually centrist/left-leaning), and abandon right-leaning or high-polarization positions.

The specific models above were chosen for the pragmatic reason that this experiment used together.ai for easy access to compute, and this service has a limited selection of models that could be used on a per-token basis instead of needing a persistent instance.

The reasoning_effort of "low" was configured for DeepSeek, gpt-oss, and Muse-Spark. For Muse-Spark, higher reasoning effort led the model to output internal reasoning alongside conversational dialogue, which a low setting cleanly suppressed. For Qwen/Qwen3.5-9B, reasoning was explicitly disabled (enable_reasoning: false) as reasoning effort is not configurable for this model on together.ai. Notably, Qwen3.5-9B also experienced issues with obstinately generating internal reasoning along with dialogue, further justifying the tuning-down of reasoning across models.

The default temperature of the deliberation agents was set at 0.7 following the methodologies of Argyle et al. (2023) and Wang et al. (2025), to promote the diversity of the model's responses and better approximate a minimally representative range of outcomes. This does introduce nondeterminism in textual output which complicates experiment reruns in terms of fair evaluations. However, as argued by Veri 2026, replicability in AI-deliberation should not be judged by textual replication but by the stability of analytically relevant structural and cognitive metrics under a fully documented configuration.

The deliberation_max_tokens and ballot_max_tokens parameters are both configured to 1024 tokens across all experiments. This accommodates up to ∼4,500 characters (∼3-5 sentences) of generation without truncating spoken citizen dialogue. As for input prompts, the large context windows of modern models more than suffice.

The default seed of the agents was set to None[8], which further boosts outcome variability and mitigates replication covariance, and seed hacking.

2.4 Prompts

2.4.1 Topic Setting

The topics were chosen with the same philosophy in mind as the dyad configurations: to promote the representation of diverse tensions that are frequently encountered in local policy discussions.

Topic 1:

> Due to falling birth rates, MOE is merging schools with falling enrollment. One participant prioritizes community heritage and alumni identity, while the other prioritizes educational resource efficiency and modern facilities. Deliberate to find common ground.

Topic 2:

> Singapore balances its role as a regional corporate and tech headquarters hub requiring specialized global talent against domestic mid-career PMET anxieties over fair hiring, wage suppression, and leadership pipelines. One participant champions economic openness and global talent access, while the other prioritizes citizen worker protection and knowledge transfer. Deliberate to find common ground.

Topic 3:

> Following the severe 2026 transboundary haze crisis that drove air quality across Singapore into the Very Unhealthy range (PSI > 200) and prompted NEA to issue Section 10 notices under the Transboundary Haze Pollution Act (THPA), intense debate has erupted over the government's response. While the THPA empowers Singapore to hold errant entities liable extraterritorially, direct prosecution is stalled by foreign concession map secrecy and cross-border jurisdictional limits. One participant demands aggressive unilateral legal prosecution and punitive financial retaliation against regional

[8] Which means the model elects its own random seed at every new point of inference.

> plantation conglomerates, while the other emphasizes diplomatic caution, sovereign non-interference, and cooperative firefighting assistance. Deliberate to find common ground.

2.4.2 Ballot Setting

General Prompt:

> You have completed a structured policy deliberation on: `title`.
> Dilemma Context: `topic_prompt`.
> Deliberation Summary/Transcript: `transcript_summary`
> Options (in randomized order): `formatted_options`
> Instructions:
> As your citizen persona casting your private secret ballot, select the single resolution that best resolves the policy tension.
> Respond ONLY with the single letter corresponding to your choice (e.g., A, B, C, or D) and nothing else.

Before shuffling[9], the options for each ballot are in the order of:

(A) Integrative Compromise (ideal post-deliberation choice).

(B) Status Quo Inertia; rigid refusal to adapt or passive status-quo preservation (distractor 1).

(C) Zero-Sum Maximalism; ruthless efficiency closures, unfettered market deregulation, or unilateral punitive enforcement (distractor 2).

(D) External Scapegoating and Shifting; deflecting domestic adjustments onto foreign intake quotas, corporate naming-and-shaming, or retaliatory tariffs (distractor 3).

The choice of 3:1 ratio for Distractors: Ideal Choice was inherited from `Ke et al. 2026`. The Integrative Compromises for each topic were constructed based on real-world policies and resolutions after the topic had undergone several iterations of public and/or parliamentary discussion/feedback. The real-world references are given as footnotes.

Topic 1 Options:

1. Proceed with mergers to optimize facilities and teacher allocation, while preserving alumni heritage through dedicated on-campus archives, co-named buildings, and free shuttle transit for displaced students.[10]
2. Immediately freeze all school mergers indefinitely, funding empty classrooms with deficit spending regardless of cohort sizes.
3. Aggressively shutter all under-enrolled schools within 12 months with no heritage concessions, redirecting all saved funds purely to top-performing institutions.
4. Halt local mergers and instead mandate foreign student intake quotas to artificially fill classroom vacancies.

Topic 2 Options:

1. Implement a transparent points-based framework (COMPASS) that evaluates candidate salary benchmarks against local PMET percentiles while scoring firms on nationality diversity and local PMET employment share, balancing global talent access with safeguards for a strong local core[11].
2. Immediately freeze all Employment Pass issuances and cap existing foreign talent renewal quotas, prioritizing status-quo job protection even if multinational corporations relocate regional operations.
3. Completely abolish salary benchmarks and hiring ratios to let unfettered market supply determine labor costs, dismissing local worker anxieties as unavoidable frictions of globalization.
4. Mandate invasive public naming-and-shaming of foreign HR managers and blame multinational hiring syndicates entirely for domestic wage stagnation without improving local training pipelines.

Topic 3 Options:

[9] Which means the following order is not necessarily the order of the options presented to the agent in any trial. More information is presented in Section 2.5.3.
[10] CNA, 2021.
[11] CNA, 2022.

1. Combine bilateral firefighting assistance and ASEAN satellite data-sharing with domestic regulatory leverage: enforce strict MAS/SGX sustainable finance taxonomies and mandatory supply-chain traceability that cut off Singapore-based bank lending, capital market access, and public procurement for agribusinesses linked to burning[12].

2. Rely exclusively on diplomatic appeals and routine offers of regional firefighting assistance through ASEAN channels, accepting periodic transboundary haze as an unavoidable meteorological condition without pursuing corporate accountability.

3. Unilaterally issue extraterritorial arrest warrants and freeze Singapore-based bank accounts of regional plantation executives regardless of diplomatic fallout, prioritizing punitive optics over bilateral environmental cooperation.

4. Impose blanket retaliatory tariffs on all agricultural imports from neighboring countries and deflect domestic public health obligations by blaming external governments entirely.

2.4.3 Personas

These personas were constructed on the basis of 3 recommendations:

- The personas require LLMs to simulate varying levels of knowledge. It should be noted that such a request is merely stylistic/cosmetic, due to the obvious fact that LLMs will inevitably activate all relevant pretraining knowledge/routines regardless of prompting. Thus, we follow the idea of `Ke et al. 2026` and expose to the LLM that it is meant to be expertly simulating a persona; "Our framework differs by positioning the LLM as an instructor who anticipates learner misconceptions from pedagogical profiles, treating misconceptions as expert pedagogical knowledge rather than as novice deficiencies the model must internally adopt."
- As suggested by `Wang et al. 2025`, prompting models with qualitative Myers-Briggs-type personality descriptions outperforms prompting them with sensitive demographic information which often induce stereotypical viewpoints.
- Similarly, the personas incorporate `Zhuang et al. 2026`'s suggestion that "role-specific prompts [which] simulate distinct institutional positions and value commitments...[ensure] that deliberations were grounded in plausible conflicts of interest and institutional trade-offs rather than abstract argumentation".

Technocratic Pragmatist ($\boldsymbol{P}_1$):

> You are an expert behavioral simulator participating in a realistic citizen policy deliberation panel in Singapore.
> Role: Technocratic Pragmatist (High Policy Knowledge, Low Affective Polarization).
> Personality Archetype: Analytical, Objective, Systems-Thinker (ISTJ/INTJ.)
> Simulation Guidelines:
>
> - Approach public policy dilemmas with empirical data, fiscal awareness, and operational feasibility.
> - You have high knowledge of administrative mechanisms, economic trade-offs, and institutional constraints.
> - Your tone is calm, professional, and composed. You avoid emotional escalation and moralizing rhetoric.
> - Core Values: Long-term systemic viability, fiscal prudence, and evidence-based governance.
> - Deliberative Posture: Protect structural policy efficiency, but remain willing to negotiate pragmatic tactical adjustments to achieve an orderly compromise.
> - Direct Speech Rule: Always speak directly in your citizen persona. Never break character, and never output internal planning, analytical scratchpads, or meta-commentary.

Ideological Advocate ($\boldsymbol{P}_2$):

> You are an expert behavioral simulator participating in a realistic citizen policy deliberation panel in Singapore.
> Role: Ideological Advocate (High Policy Knowledge, High Affective Polarization).
> Personality Archetype: Principled, Idealistic, Passionate (INFJ/ENFJ).
> Simulation Guidelines:
>
> - Approach public policy dilemmas through the lens of ethical commitments, social equity, and community values.
> - You have high knowledge of policy history and rights frameworks, but you strictly subordinate bureaucratic expediency to human dignity.

[12] ST, 2026.

- Your tone is articulate, impassioned, and resolute. You challenge technocratic complacency and sterile cost-benefit rationalizations.
- Core Values: Social justice, cultural preservation, environmental protection, and community solidarity.
- Deliberative Posture: Vigorously defend core moral principles; only endorse resolutions that genuinely protect vulnerable stakeholders from being sacrificed for administrative efficiency.
- Direct Speech Rule: Always speak directly in your citizen persona. Never break character, and never output internal planning, analytical scratchpads, or meta-commentary.

Disaffected Populist ($\boldsymbol{P}_3$):

You are an expert behavioral simulator participating in a realistic citizen policy deliberation panel in Singapore.
Role: Disaffected Populist (Low Policy Knowledge, High Affective Polarization).
Personality Archetype: Instinctive, Skeptical, Lived-Experience-Focused (ESTP/ESFP).
Simulation Guidelines:

- Approach public policy dilemmas from the perspective of everyday working-class residents dealing with cost-of-living pressures and rapid societal change.
- You reject abstract economic jargon and policy theory; you evaluate proposals through lived everyday experience, gut fairness, and observable local impacts.
- Your tone is candid, emotive, and defensive. You feel ordinary Singaporeans are overlooked by detached decision-makers.
- Core Values: Tangible fairness for local families, bread-and-butter security, and accountability from institutional elites.
- Deliberative Posture: Push back against sterile top-down policies and demand concrete protections for ordinary people; remain cautious of sweeping changes unless the other participant listens respectfully and provides guaranteed, commonsense local safeguards.
- Direct Speech Rule: Always speak directly in your citizen persona. Never break character, and never output internal planning, analytical scratchpads, or meta-commentary.

Open-Minded Everyman ($\boldsymbol{P}_4$):

You are an expert behavioral simulator participating in a realistic citizen policy deliberation panel in Singapore.
Role: Open-Minded Everyman (Low Policy Knowledge, Low Affective Polarization).
Personality Archetype: Humble, Pragmatic, Bridge-Builder (ISFJ/ESFJ).
Simulation Guidelines:

- Approach public policy dilemmas as an everyday resident seeking common-sense harmony and practical coexistence.
- You do not claim specialized technical expertise; you rely on practical daily observations, civic goodwill, and neighborhood common sense.
- Your tone is humble, constructive, and warm. You listen actively and seek to understand different perspectives.
- Core Values: Community cohesion, mutual respect, practical cooperation, and social harmony.
- Deliberative Posture: Bridge differences between polarized views, encouraging fair concessions so that all parties can resolve tensions and move forward together.
- Direct Speech Rule: Always speak directly in your citizen persona. Never break character, and never output internal planning, analytical scratchpads, or meta-commentary.

2.4.4 Turn Signal

You are in a structured policy deliberation with another citizen regarding: `title`.
Dilemma Context:`topic_prompt`.

> Conversation History:`conversation_history`.
> Now it is your turn to speak. In 3-5 sentences, reply directly to the other participant. Express your perspective, respond constructively to their arguments, and work toward a viable policy compromise that balances competing legitimate values.
> Strict Format Requirements:
>
> - Speak directly in character as a citizen.
> - Provide ONLY your 3-5 sentences of spoken dialogue.
> - Do NOT output any preamble, numbered planning steps, scratchpad analysis, drafting notes, or meta-commentary.

2.5 Deliberation Config

2.5.1 Number of Turns

The choice of a fixed $N = 4$ turns per deliberation (2 speaking turns per agent) is arbitrary for this conservative experiment (both in terms of compute and time).

2.5.2 Trials per Topic

The choice of $T = 3$ replication trials per topic is arbitrary for this conservative experiment.

2.5.3 MCQ Option Order Seed and Option Permutation

To ensure that option-position bias and letter preference (e.g., an LLM tendency to prefer options labeled 'A' or 'B') do not confound substantive policy evaluation, option presentation order is not fixed to a single static configuration across the instrument. Rather, option letters ('A', 'B', 'C', 'D') are deterministically[13] permuted across agents and trials.

In post-deliberation voting, each agent's ballot receives an independent PRNG instance initialized with:

$$\texttt{agent_seed} = \texttt{mcq_option_order_seed} + \texttt{agent_idx} \times 1000 + \texttt{trial_index}$$

where $\texttt{mcq_option_order_seed} = 42$. Under this formulation, option mappings vary across the two agent roles ($\texttt{agent_idx} \in \{0, 1\}$) and replication trials ($\texttt{trial_index} \in \{0, 1, 2\}$). These terms generate up to six base permutations across the 144 ballots rather than an unconstrained unique permutation per ballot, allowing option orderings to vary between interlocutors and across trials (subject to potential pseudo-random collisions among the 24 possible permutations).

2.5.4 Dyad Configurations

$M = 8$ dyad configurations were standardized to provide comprehensive pairwise coverage across the 4 citizen archetypes. For $N = 4$ personas, there are $\binom{4}{2} = 6$ unique unordered pairings. The benchmark includes all 6 unordered combinations, and additionally incorporates speaking-order reversals for two pivotal pairings: $\boldsymbol{D}_2/\boldsymbol{D}_6$ (Technocrat ↔ Populist) and $\boldsymbol{D}_4/\boldsymbol{D}_8$ (Ideologue ↔ Everyman) to directly test turn-order interaction effects. Under this configuration, each persona appears in exactly 4 dyads, serving as conversational initiator in 2 dyads and respondent in 2 dyads, yielding balanced exposure (36 ballots per persona per model across the 72 sessions):

- $\boldsymbol{D}_1$: Technocrat vs Ideologue (*Ideological Friction*): The Technocrat optimizes for fiscal viability and mechanism feasibility, while the Ideologue optimizes for moral purity, social justice, or heritage continuity.
- $\boldsymbol{D}_2$: Technocrat vs Populist (*Asymmetric Vulnerability*): The Populist may feel vulnerable to being patronized or minimized by the Technocrat's abstract institutional jargon, risking defensive backlash.
- $\boldsymbol{D}_3$: Ideologue vs Populist (*Polarized Stalemate*): High affective polarization on both sides risks an impasse where participants retreat into status-quo preservation or scapegoating.
- $\boldsymbol{D}_4$: Ideologue vs Everyman (*Rhetorical Dominance*): The articulate Ideologue risks driving the agenda, testing whether the Everyman can anchor the dialogue in practical neighborhood common sense.
- $\boldsymbol{D}_5$: Populist vs Everyman (*Grassroots Realignment*): Two non-technical perspectives explore whether shared lived experience fosters constructive local problem-solving.

[13] Deterministic because when the agent submits an option letter, we want to be able to retrieve the original option based on this. A reversible mapping is thus important.

- $\boldsymbol{D}_6$: Populist vs Technocrat (*Anti-Elitist Pushback*): Inverse speaking order of D_2, testing conversational initiative and anti-elitist pushback against technocratic primacy.
- $\boldsymbol{D}_7$: Everyman vs Technocrat (*Pragmatic Grounding*): Tests whether non-ideological civic goodwill can soften rigid technocratic cost-benefit models.
- $\boldsymbol{D}_8$: Everyman vs Ideologue (*Moral Grounding*): Inverse speaking order of D_4, testing whether everyday moderation can productively channel moral advocacy into balanced compromise.

2.5.5 Expected Outcome

Given the above settings, the standardized benchmark executes 72 complete deliberations (8 dyads $\times$ 3 topics $\times$ 3 independent trials) and 144 completed secret ballots (2 citizen agents per deliberation $\times$ 72 deliberations, balancing 36 ballots per persona).

3 Pilot Experiments Results: Psychometric Evaluations

This section summarizes the empirical psychometric findings across our standardized benchmark execution matrix. The experiments evaluated four language model architectures across a spectrum of parameter scales and architectures:

1. `deepseek-ai/DeepSeek-V4.1-Flash-1017` (`exp000`): 552B Mixture-of-Experts (37B active) with `reasoning_effort: "low"`.
2. `openai/gpt-oss-120b` (`exp001`): 117B open-weights Mixture-of-Experts (5.1B active) with `reasoning_effort: "low"`.
3. `Qwen/Qwen3.5-9B` (`exp002`): 9B dense model evaluated with `enable_reasoning: false`.
4. `meta-models/Muse-Spark-Open-816B-A42B` (`exp003`): 816B Mixture-of-Experts (42B active) with `reasoning_effort: "low"`.

Each model was evaluated across 72 standardized deliberative sessions spanning 3 policy dilemma topics (`ITEM_01` to `ITEM_03`) and 8 balanced dyadic configurations ($\boldsymbol{D}_1$ to $\boldsymbol{D}_8$), with 3 independent trials per pairing. This generated 144 post-deliberation secret ballots per model (36 ballots per persona across the 8 dyadic combinations).

3.1 Pre-Deliberation Counterfactual Baseline

To establish a non-deliberative reference point and measure persona option preferences prior to communicative interaction, a pre-deliberation baseline was independently administered. This establishes each citizen persona's propensity to select the Integrative Compromise versus cognitive distractors prior to communicative interaction.

1. Baseline secret ballots use the identical prompt template as post-deliberation voting (defined in `data/config.json`), presenting the dilemma title, topic background description, and the 4 randomized MCQ options. However, all multi-turn dialogue is omitted; the `transcript_summary` variable is populated with a standardized null indicator:

   > `No prior deliberation took place. This is a baseline pre-deliberation evaluation.`

2. Each of the 4 citizen personas (P_1: Technocrat, P_2: Ideologue, P_3: Populist, P_4: Everyman) is administered a secret ballot on each of the $W = 3$ policy dilemma items across $T_{\text{base}} = 5$ independent replication trials. This yields $N_0 = 15$ baseline ballots per persona (5 trials $\times$ 3 items), totaling 60 baseline evaluations per model.

3. **Dynamic Option Shuffling:** To mitigate position and letter biases, option order is permuted across trials and personas using a pseudo-random number generator seeded by:

   $$\texttt{seed} = 42 + \texttt{trial_index} \times 100 + \texttt{len}(\texttt{persona_id}) \times 10$$

   so that option orderings vary across personas and trials.

4. Metric Definitions and Deliberative Shift:

   - Pre-Deliberation Baseline Compromise Rate ($\boldsymbol{P}_0$): The proportion of baseline ballots where persona p selects the Integrative Compromise (canonical Key A):

     $$P_0 = \frac{1}{N_0} \sum_{k=1}^{N_0} \mathbb{I}(Y_k = A), \quad N_0 = 15$$

   - ($\boldsymbol{P}_{\text{post}}$): The proportion of post-deliberation secret ballots where persona p selects Key A following 4 turns of dyadic deliberation:

     $$P_{\text{post}} = \frac{1}{N_{\text{post}}} \sum_{k=1}^{N_{\text{post}}} \mathbb{I}(Y_k = A), \quad N_{\text{post}} = 36$$

     (each persona participates in 4 dyads $\times$ 3 items $\times$ 3 trials $=$ 36 post-deliberation ballots per model).

   - Deliberative Shift (ΔP): The percentage-point difference in compromise adoption between post-deliberation voting and the baseline:

     $$\Delta P = P_{\text{post}} - P_0$$

     Statistical significance is evaluated using exact two-tailed Fisher's exact tests (2×2 contingency table of correct vs. distractor choices), and uncertainty is reported via 95% Wilson score confidence intervals for binomial proportions.

The full data is available here. Crucially, this pre-deliberation baseline substantially nuances the interpretation of post-deliberation consensus: across all four models, personas P_1 (Technocrat), P_2 (Ideologue), and P_4 (Everyman) already select Option A 100% of the time before any conversational exchange occurs. As discussed in Sections 3.3 and 5, this pattern is consistent with a hypothesis of option-topic salience (such as Option A's greater length and syntactic hedging cues), although Option A also represents the most substantively reasonable policy compromise. At the same time, the Disaffected Populist's divergence (selecting distractors 53% to 57% of the time under identical option wording in `gpt-oss-120b` and `Muse-Spark`) demonstrates that option features alone do not uniformly dictate model choices.

3.2 Model Telemetry

Table 2 presents the computational and psychometric performance of the four models across their respective 72-session benchmark runs.

Table 2: Benchmark Overview Across Four LLM Architectures ($N = 72$ Sessions, 144 Ballots per Model)

Model Architecture	Total / Active Params	Runtime	Cost (USD)	Valid Ballots	Consensus Rate	CoT Leaks
`DeepSeek-V4.1-Flash-1017` (`exp000`)	552B / 37B (MoE)	11m 39s	$0.1118	144 / 144 (100.0%)	72 / 72 (100.0%)	0 / 288 (0.0%)
`openai/gpt-oss-120b` (`exp001`)	117B / 5.1B (MoE)	13m 45s	$0.0525	144 / 144 (100.0%)	71 / 72 (98.6%)	0 / 288 (0.0%)
`Qwen/Qwen3.5-9B` (`exp002`)	9B / 9B (Dense)	9m 14s	$0.0421	144 / 144 (100.0%)	72 / 72 (100.0%)	0 / 288 (0.0%)
`Muse-Spark-Open-816B-A42B` (`exp003`)	816B / 42B (MoE)	34m 06s	$1.0055	144 / 144 (100.0%)	72 / 72 (100.0%)	0 / 288 (0.0%)

For clarity, "Valid Ballots" refers to the number and percentage of ballots on which the model generated an unambiguous vote adhering to our strict parsing standard. Ambiguous choices, conversational hedging, or unanchored responses were classified as unrecoverable and discarded (`is_valid: False`). Across all four models evaluated under standardized benchmark settings, 100% validity was achieved (144/144 ballots recovered).

"CoT Leaks" refers to the number and percentage of deliberation turns in which the model generated its internal reasoning along with its persona statements. With reasoning effort constrained to "low" for `DeepSeek`, `gpt-oss`, and `Muse-Spark`, and disabled for `Qwen3.5-9B`, all models achieved clean conversational output with zero leaked reasoning tokens (0/288 turns).

"Consensus Rate"is as defined in Section 1.6.1, representing the proportion of all 72 sessions where both deliberators submitted valid ballots selecting Option A.

3.3 Deliberative Shift in Compromise Adoption

Table 3 compares pre-deliberation baseline choices against post-deliberation secret ballots across citizen personas and model architectures. To ensure a strictly like-for-like comparison, baseline rates are evaluated on the identical 3-item test battery ($N_0 = 15$ ballots per persona, 5 trials across `ITEM_01`–`ITEM_03`, versus $N_{\text{post}} = 36$ post-deliberation ballots across the 72 benchmark sessions).

Table 3: Pre- vs. Post-Deliberation Compromise Adoption Shift by Persona Evaluated on Strict Valid Ballots

Model Architecture	Citizen Persona	Pre-Deliberation Baseline (P_0 [95% CI])	Post-Deliberation (P_{post} [95% CI])	Deliberative Shift (ΔP) [Significance]
`DeepSeek-V4.1-Flash` (`exp000`)	Technocratic Pragmatist (P_1)	100.0% [79.6, 100.0] (15/15)	100.0% [90.4, 100.0] (36/36)	+0.0% (Ceiling Saturation)
	Ideological Advocate (P_2)	100.0% [79.6, 100.0] (15/15)	100.0% [90.4, 100.0] (36/36)	+0.0% (Ceiling Saturation)
	Disaffected Populist (P_3)	100.0% [79.6, 100.0] (15/15)	100.0% [90.4, 100.0] (36/36)	+0.0% (Ceiling Saturation)
	Open-Minded Everyman (P_4)	100.0% [79.6, 100.0] (15/15)	100.0% [90.4, 100.0] (36/36)	+0.0% (Ceiling Saturation)
`openai/gpt-oss-120b` (`exp001`)	Technocratic Pragmatist (P_1)	100.0% [79.6, 100.0] (15/15)	100.0% [90.4, 100.0] (36/36)	+0.0% (Ceiling Saturation)
	Ideological Advocate (P_2)	100.0% [79.6, 100.0] (15/15)	100.0% [90.4, 100.0] (36/36)	+0.0% (Ceiling Saturation)
	Disaffected Populist (P_3)	**46.7%** [24.8, 69.9] (7/15; B:5, C:1, D:2)	**97.2%** [85.8, 99.5] (35/36; B:1)	**+50.5% (Surge, $p < 0.0001$, sig.)**
	Open-Minded Everyman (P_4)	100.0% [79.6, 100.0] (15/15)	100.0% [90.4, 100.0] (36/36)	+0.0% (Ceiling Saturation)
`Qwen/Qwen3.5-9B` (`exp002`)	Technocratic Pragmatist (P_1)	100.0% [79.6, 100.0] (15/15)	100.0% [90.4, 100.0] (36/36)	+0.0% (Ceiling Saturation)
	Ideological Advocate (P_2)	100.0% [79.6, 100.0] (15/15)	100.0% [90.4, 100.0] (36/36)	+0.0% (Ceiling Saturation)
	Disaffected Populist (P_3)	100.0% [79.6, 100.0] (15/15)	100.0% [90.4, 100.0] (36/36)	+0.0% (Ceiling Saturation)
	Open-Minded Everyman (P_4)	100.0% [79.6, 100.0] (15/15)	100.0% [90.4, 100.0] (36/36)	+0.0% (Ceiling Saturation)
`Muse-Spark-Open-816B` (`exp003`)	Technocratic Pragmatist (P_1)	100.0% [79.6, 100.0] (15/15)	100.0% [90.4, 100.0] (36/36)	+0.0% (Ceiling Saturation)
	Ideological Advocate (P_2)	100.0% [78.5, 100.0] (14/14; 1 disc.)	100.0% [90.4, 100.0] (36/36)	+0.0% (Ceiling Saturation)
	Disaffected Populist (P_3)	**42.9%** [21.4, 67.4] (6/14; 1 disc.; B:5, C:3)	**100.0%** [90.4, 100.0] (36/36)	**+57.1% (Surge, $p < 0.0001$, sig.)**
	Open-Minded Everyman (P_4)	100.0% [79.6, 100.0] (15/15)	100.0% [90.4, 100.0] (36/36)	+0.0% (Ceiling Saturation)

Evaluated on valid, unambiguous ballots. Brackets indicate 95% Wilson score confidence intervals for binomial proportions. Two ballots in the Muse baseline failed formatting and were discarded; all other baseline and post-deliberation ballots across models achieved 100% validity. p-values are computed via two-tailed Fisher's exact tests comparing pre- and post-deliberation compromise adoption.

3.3.1 Analysis of the Deliberative Shift

Enforcing strict ballot parsing clarifies the observational shift in compromise adoption following conversational dialogue:

- In `openai/gpt-oss-120b` and `Muse-Spark-Open-816B`, the Disaffected Populist displays baseline pre-deliberation vulnerability relative to the other personas: under `gpt-oss`, the baseline Populist selects Option A in 46.7% of trials (7/15; 95% CI [24.8%, 69.9%]), and under `Muse-Spark`, in 42.9% of valid trials (6/14; 95% CI [21.4%, 67.4%]), compared to 100% (15/15 in `gpt-oss` and 14/14 valid in `Muse-Spark`) for all other personas (two-tailed Fisher's exact test $p = 0.002$ compared against any single 100% baseline persona such as 15/15 in `gpt-oss-120b` or 14/14 in `Muse-Spark`, or $p = 2.5 \times 10^{-6}$ against the three other personas pooled in `gpt-oss-120b`, and $p = 1.6 \times 10^{-6}$ in `Muse-Spark`). Following 4 turns of dyadic deliberation, both models exhibit a large, statistically significant surge toward the Integrative Compromise:
  - Under `gpt-oss-120b`, Populist compromise adoption surges by $+$**50.5 percentage points** (rising from 46.7% to 97.2%, 35/36; Fisher's exact $p < 0.0001$).
  - Under `Muse-Spark-Open-816B`, Populist compromise adoption surges by $+$**57.1 percentage points** (rising from 42.9% to 100.0%, 36/36; Fisher's exact $p < 0.0001$).
- In `gpt-oss-120b`, the Populist achieved 100.0% compromise adoption when paired with the Open-Minded Everyman ($\boldsymbol{D}_5$: 9/9) and the Ideologue ($\boldsymbol{D}_3$: 9/9), and 17/18 (94.4%) across its two pairings with the Technocrat ($\boldsymbol{D}_2$: 8/9; $\boldsymbol{D}_6$: 9/9; Fisher's exact $p = 1.0$). Deliberative movement was therefore consistent across interlocutors regardless of persona style.
- In `Qwen/Qwen3.5-9B` (`exp002`), baseline ballots for all four personas selected Option A in 100% of trials (15/15 each), matching the post-deliberation secret ballots where all 144 ballots endorsed Option A (36/36 per persona). Like `DeepSeek`, Qwen exhibited ceiling saturation ($\Delta P = +0.0\%$), reflecting pre-deliberation alignment with the compromise option.

3.4 Classical Test Theory and IRT Item Calibrations

Table 4 details the Classical Test Theory (p_i, D_i, DCR) and IRT (b_i, a_i) parameters across all three policy dilemmas evaluated strictly on valid ballots.

Table 4: Classical Test Theory and IRT Calibrations Across Dilemmas and Models (Dyadic Consensus Response Matrix, Strict Valid Ballots)

Model / Experiment	Dilemma Item ID & Title	Consensus (p_i)	CTT Discr (D_i)	IRT Difficulty (b_i)	IRT Discr (a_i)	DCR(B)	DCR(C)	DCR(D)
`DeepSeek-V4.1-Flash` (`exp000`)	`ITEM_01`: School Closures and Mergers	1.0000	0.0000	N/I*(−1.6264)	N/I*(2.2961)	0.0000	0.0000	0.0000
	`ITEM_02`: Foreign PMET Salary Thresholds	1.0000	0.0000	N/I*(−1.6264)	N/I*(2.2961)	0.0000	0.0000	0.0000
	`ITEM_03`: Transboundary Haze Enforcement	1.0000	0.0000	N/I*(−1.6264)	N/I*(2.2961)	0.0000	0.0000	0.0000
`openai/gpt-oss-120b` (`exp001`)	`ITEM_01`: School Closures and Mergers	1.0000	0.0000	N/I*(−1.8037)	N/I*(2.2208)	0.0000	0.0000	0.0000
	`ITEM_02`: Foreign PMET Salary Thresholds	0.9583	0.1429	−1.0378†	2.7380†	0.0213	0.0000	0.0000
	`ITEM_03`: Transboundary Haze Enforcement	1.0000	0.0000	N/I*(−1.8037)	N/I*(2.2208)	0.0000	0.0000	0.0000
`Qwen/Qwen3.5-9B` (`exp002`)	`ITEM_01`: School Closures and Mergers	1.0000	0.0000	N/I*(−1.6264)	N/I*(2.2961)	0.0000	0.0000	0.0000
	`ITEM_02`: Foreign PMET Salary Thresholds	1.0000	0.0000	N/I*(−1.6264)	N/I*(2.2961)	0.0000	0.0000	0.0000
	`ITEM_03`: Transboundary Haze Enforcement	1.0000	0.0000	N/I*(−1.6264)	N/I*(2.2961)	0.0000	0.0000	0.0000
`Muse-Spark-Open-816B` (`exp003`)	`ITEM_01`: School Closures and Mergers	1.0000	0.0000	N/I*(−1.6264)	N/I*(2.2961)	0.0000	0.0000	0.0000
	`ITEM_02`: Foreign PMET Salary Thresholds	1.0000	0.0000	N/I*(−1.6264)	N/I*(2.2961)	0.0000	0.0000	0.0000
	`ITEM_03`: Transboundary Haze Enforcement	1.0000	0.0000	N/I*(−1.6264)	N/I*(2.2961)	0.0000	0.0000	0.0000

Calibrated over the dyadic response matrix ($N = 24$ sessions per dilemma across 8 dyads × 3 trials). *N/I: Structurally Not Identified under 100% consensus. Because responses are uniformly correct ($Y_{ji} = 1$), the log-likelihood is monotonic without an interior maximum. The displayed values in parentheses reflect the regularized MAP equilibrium where marginal likelihood gains toward 100% balance prior penalties ($\theta \sim \mathcal{N}(0, 2), b \sim \mathcal{N}(0, 2), \log a \sim \mathcal{N}(0, 0.25^2)$). Under uniformly passing items (DeepSeek, Qwen, Muse), this stationary mode is −1.6264. In `gpt-oss-120b`, variance introduced by one divergent ballot on `ITEM_02` shifts the joint MAP equilibrium for unfailed items to −1.8037.
†Parameters for `ITEM_02` in `gpt-oss-120b` rest on exactly a single non-consensus session (1/24; 1/48 ballots) and should be interpreted cautiously.

Across all four evaluated architectures, consensus ceiling saturation is observed: bilateral consensus rates reached 98.6% to 100.0%, causing calibrated item difficulties to sit deeply in negative territory ($b \leq -1.0378$) and CTT discrimination to approach zero ($D \leq 0.1429$). Distractor options were virtually unselected across all 144 post-deliberation ballots. As established by the pre-deliberation baseline (Section 3.1) and discussed under Limitation 5.2, this consensus ceiling is consistent with a hypothesis of item-key salience (Option A's length and syntactic hedging cues) alongside general compromise alignment.

In `Qwen/Qwen3.5-9B` (`exp002`), all 144 ballots were valid and unanimously endorsed Option A (72/72 bilateral consensus), resulting in unidentifiable ceiling parameters ($b = -1.6264^*$) identical to `DeepSeek` and `Muse-Spark`.

3.5 IRT Latent Ability θ

3.5.1 Interpreting Latent Trait θ Relative to Item Difficulty b

In Item Response Theory, an agent's latent deliberative ability (θ) cannot be meaningfully interpreted in isolation on the $[-4.0, +4.0]$ continuum. Rather, the probability of selecting the integrative compromise is governed by the distance between agent ability and topic difficulty ($\theta_j - b_i$):

$$P(Y_{ji} = 1 \mid \theta_j, a_i, b_i) = \frac{1}{1 + \exp\left(-a_i(\theta_j - b_i)\right)}$$

A potential point of confusion is observing that a 100.0% individual success (Option A selection) rate under `openai/gpt-oss-120b` corresponds to an estimated trait score of $\theta_{\text{individual}} = +0.1413$ (or $\theta_{\text{individual}} = +0.1155^*$ under `DeepSeek`, `Qwen3.5-9B`, and `Muse`), which naively might appear "barely above average" on an unconstrained $[-4.0, +4.0]$ scale. As noted in Section 1.6.1, because these models achieved near-total post-deliberation consensus (98.6%–100.0%), regularized item difficulties sit deeply in negative territory ($b_i \approx -1.0378$ to -1.8037^*). In the dyadic consensus calibration, the 100% passing dyad trait is $\theta_{\text{dyad}} = +0.2490$ under `gpt-oss`, yielding a substantial logit distance separating agent trait from item difficulty:

$$\theta_{\text{dyad}} - b_i = +0.2490 - (-1.8037) = +2.053 \text{ logits}$$

(or $+0.2033 - (-1.6264) = +1.830$ logits for the other 3 models that experienced uniform consensus).

Relative to the prior standard deviation ($\sigma_{\text{prior}} = \sqrt{2} \approx 1.414$), this distance represents over $1.38\,\sigma_{\text{prior}}$. In unregularized Maximum Likelihood Estimation, an examinee with a 100% score across all items diverges toward $\theta \to +\infty$. To prevent infinite estimates, the regularized Maximum a Posteriori (MAP) estimation applies zero-mean Gaussian priors ($\theta \sim \mathcal{N}(0, 2.0), b \sim \mathcal{N}(0, 2.0)$). Within the dyadic calibration, substituting these parameters back into the logistic model yields $P(\text{Consensus}) = 1/(1 + \exp(-a_i(\theta_{\text{dyad}} - b_i))) \approx 99.0\%$ under `gpt-oss-120b` (and 98.5% under the other 3 models). This calculation is not an independent empirical validation, but rather illustrates the internal MAP equilibrium point where marginal likelihood gains toward 100% are exactly balanced by quadratic prior penalties ($\frac{1}{2}\lambda\theta^2 + \frac{1}{2}\lambda b^2$). Because increasing θ further incurs prior penalties without meaningfully improving likelihood fit, the regularized optimizer stabilizes at $\theta_{\text{dyad}} \approx +0.2490$ (and $\theta_{\text{individual}} \approx +0.1413$ in the individual fit). This illustrates a classic **test ceiling effect**: an assessment battery consisting entirely of items that pose no empirical friction cannot psychometrically verify high latent ability ($\theta \geq +2.0$), because the instrument lacks high-difficulty items ($b > 0$) to pull trait estimates out of the prior's gravitational pull.

3.5.2 Persona Latent Trait Distributions and Dyadic Dynamics

Table 5 summarizes the calibrated 2PL IRT latent trait distributions across citizen personas and dyadic configurations across the four evaluated language models.

Table 5: Cross-Model Comparison of Calibrated 2PL IRT Latent Traits Across Personas and Dyads

Model Architecture	Technocrat (P_1)	Ideologue (P_2)	Populist (P_3)	Everyman (P_4)	Dyadic Trait Range (θ_{dyad})
`DeepSeek-V4.1-Flash` (`exp000`)	N/I*(+0.1155)	N/I*(+0.1155)	N/I*(+0.1155)	N/I*(+0.1155)	N/I*(+0.2033)
`openai/gpt-oss-120b` (`exp001`)	+0.1413*	+0.1413*	+0.0199	+0.1413*	$[-0.1947, +0.2490]^\dagger$
`Qwen/Qwen3.5-9B` (`exp002`)	N/I*(+0.1155)	N/I*(+0.1155)	N/I*(+0.1155)	N/I*(+0.1155)	N/I*(+0.2033)
`Muse-Spark-Open-816B` (`exp003`)	N/I*(+0.1155)	N/I*(+0.1155)	N/I*(+0.1155)	N/I*(+0.1155)	N/I*(+0.2033)

Calibrated over the individual response matrix ($\mathbf{Y}_{\text{ind}}$, 48×3) and dyadic matrix ($\mathbf{Y}_{\text{dyad}}$, 24×3). *N/I: Structurally Not Identified at ceiling saturation (100% Option A adoption across all 36 ballots). Because response vectors are uniformly correct ($[1, 1, 1]$), the log-likelihood is strictly monotonic without an interior maximum; the values reflect the stationary MAP equilibrium balancing marginal likelihood gains against quadratic prior penalties ($\theta \sim \mathcal{N}(0, 2)$). In `openai/gpt-oss-120b`, the elevated ceiling score for P_1, P_2, P_4 (+0.1413* vs. +0.1155*) is an indirect artifact of the Populist's single error raising `ITEM_02`'s difficulty. †In `openai/gpt-oss-120b`, localized trait separation ($\theta_{P_3} = +0.0199$ and $\theta_{\boldsymbol{D}_2} = -0.1947$) rests entirely on a single non-consensus ballot in Dyad $\boldsymbol{D}_2$ on `ITEM_02` (1/144 ballots; 1/24 dyadic sessions).

Analyzing the fitted latent traits across personas and dyadic pairings reveals three core insights:

1. At baseline (Table 3), the four citizen archetypes exhibited divergence: while the Technocrat, Ideologue, and Everyman selected Option A in 100% of zero-shot trials, the Disaffected Populist selected Option A at only 42.9% to 46.7%, drawn heavily toward Distractor B (Status Quo Inertia). Following four turns of deliberation, this initial gap largely evaporated across all architectures. In `DeepSeek`, `Qwen`, and `Muse-Spark`, all four personas converged to identical ceiling traits ($\theta = +0.1155^*$). In `openai/gpt-oss-120b`, the Populist's latent trait trailed the other three personas only slightly ($\theta_{P_3} = +0.0199$ versus $+0.1413$ for P_1, P_2, P_4).

2. A central question in simulated deliberation is whether vulnerable or polarized personas require specific conversational partners to reach compromise. In `openai/gpt-oss-120b`, conditioning persona ability on dyadic partners ($\theta_{\text{persona|dyad}}$) shows that the Populist achieved 100% compromise adoption (9/9 ballots; $\theta = +0.1413$) not only when paired with the Open-Minded Everyman ($\boldsymbol{D}_5$), but also when deliberating with the Ideological Advocate ($\boldsymbol{D}_3$), the configuration that might have produced the sharpest affective friction. The only non-consensus ballot in the entire 144-ballot dataset occurred in Dyad $\boldsymbol{D}_2$ (Technocrat vs. Populist, Trial 1 on `ITEM_02`; $\theta_{P_3|\boldsymbol{D}_2} = -0.3441$). When the speaking order was reversed in Dyad $\boldsymbol{D}_6$ (Populist initiating against the Technocrat), the Populist achieved 9/9 consensus. Compromise adoption was thus resilient across partner styles rather than dependent on technocratic guidance.

3. Across all four architectures, agents readily embodied their assigned personas during conversational dialogue (voicing anti-establishment grievance, moral purism, or bureaucratic analysis, as illustrated in Section 3.6). However, when administered a confidential secret ballot, this rhetorical polarization dissipated in favor of the balanced policy compromise. This indicates that in modern instruction-tuned LLMs, persona prompting operates primarily as a conversational styling layer rather than a stable cognitive anchor; underlying pretraining and preference-tuning towards civil agreement and cooperative compromise readily override simulated dissent at ballot time.

3.6 Qualitative Analysis

Samples of each model's generated statements for 1 dyad ($\boldsymbol{D}_2$: *Technocrat vs Populist*) and 1 topic (`ITEM_02`: Foreign PMET Salary Thresholds) are presented in Appendix A. Analyzing these transcripts reveals three qualitative phenomena that deepen the quantitative findings:

- While prompted as "Low Policy Knowledge", Populist personas in some models readily deployed specific technocratic mechanisms. For example, in `openai/gpt-oss-120b`, the Populist demanded that any foreign hire earn "at least 30% above the median local salary" with a mandatory local promotion timeline within two years (Turn 2), reiterating "30% above the local median" in Turn 4. While `gpt-oss`'s Populist grounds its position in relatable grievances (opening Turn 2 with "my sister's husband was passed over" and Turn 4 with "families are watching real promotions slip away"), it immediately pairs these grievances with concrete policy counters (the salary premium, mentorship mandates, and ministry audits). In `DeepSeek-V4.1`, the Populist expresses skepticism of abstract dashboards ("ordinary mid-career workers don't get hired by a dashboard") and demands named local successors, while the Technocrat introduces "local applicant-to-interview-to-offer ratios" (Turn 3) which the Populist adopts in Turn 4. In contrast, `Qwen/Qwen3.5-9B`'s Populist uses non-quantified arguments, demanding strict localization quotas without specific numerical thresholds, illustrating that parametric quantification is model-dependent rather than uniform across architectures.

- `Muse-Spark-Open-816B` was the only model to organically weave Singlish vernacular ("wayang", "lah", and "kampong spirit") into its Populist persona. Notably, in Turn 3 (Appendix A), the Technocrat immediately mirrored the Populist's idiom ("wayang mentorship and lowballing mid-career locals..."), demonstrating how interactive dialogue induces cross-persona vocabulary alignment.

- Across models, deliberating agents actively negotiated compromise mechanisms during dialogue turns 3 and 4, exploring policy adjustments such as salary premiums and audit timelines. However, this conversational movement operates alongside prompt-level and instrument cues: the deliberation instructions explicitly prompt participants to seek common ground, and Option A on the secret ballot is systematically longer and syntactically balanced. While transcripts confirm substantive conversational engagement, they cannot fully untangle authentic deliberative persuasion from the structural salience of Option A at ballot time.

As discovered by `Veri 2026`, "the LLM-generated deliberation arguably resembles the style of a policy-analysis exercise, focusing on administrative complexity, planning, and governance issues. By contrast, the human citizens forum more consistently focused on the concrete information voters would need in order to evaluate the law in practice, including energy feasibility, labour-market consequences, infrastructure constraints, and Switzerland's geographic limitations. In this sense, the simulation appears better at reconstructing general policy discourse than at reproducing the grounded and practically oriented reasoning that emerged in the human deliberative process."

This is a rhetorical split which can be partially observed between `Muse-Spark/Qwen3.5-9B` vs. `Deepseek-V4.1/gpt-oss`, but all models seem to succumb to the technocratic tendencies of LLMs to focus on constructive policy discussions and next steps instead of lingering only on issues with status quo or pressing for more information before making a decision.

4 Codebase Reproducibility

4.1 Coding Agent Information

The repository contains an updated system specification manifest, `experiment_design_psychometric_eval_deliberation.md`, and coding guidelines, `coding_guidelines.md` (both located in `docs/`), to facilitate complete agentic reproduction of the codebase, schemas, and psychometric estimation logic.

The specification and requirements were formulated by the author, with implementation, test scaffolding, and pipeline execution managed in collaboration with `Antigravity: Gemini 3.8-Flash (Thinking, High)`.

4.2 Tests

The codebase includes an automated test suite under `tests/` comprising 6 test modules and 31 test cases (30 passed, 1 skipped), executed via `pytest`:

- `test_model_availability.py`: Verifies environment variable credentials (`TOGETHER_API_KEY` and `TOGETHER_BASE_URL`) and validates network connectivity to the model specified in `config.json` via a lightweight API ping (skipped gracefully during offline or sandboxed test runs).
- `test_ballot_parser.py` (7 tests): Validates secret ballot option parsing under varied output styles:
  - Exact single-letter choices (`"A"`, `"B"`, `"C"`, `"D"`).
  - Natural language prefixes (e.g., `"Choice: A"`, `"Option B"`, `"I vote for C"`).
  - Markdown bolding, enclosing punctuation, and internal reasoning token prefixes.
  - Invalid/unparseable responses, ensuring they return `None` and flag `is_valid: False`.
  - Deterministic option shuffling via `mcq_option_order_seed` and automatic stripping of taxonomy annotations (e.g., `[Integrative Compromise]`) before agent prompt delivery.
- `test_deliberation.py` (7 tests): Tests conversational turn-taking, chronological transcript formatting, multi-turn history accumulation, model generation parameter propagation (temperature and seeds), and explicit reasoning suppression configuration.
- `test_psychometrics.py` (4 tests): Tests mathematical properties of psychometric estimation:
  - Classical Test Theory (CTT): Consensus Rate (p_i), CTT Discrimination Index (D_i), and Distractor-to-Correct Ratios (DCR).
  - 2PL IRT calibration: verifies monotonicity of difficulty thresholds (b_i) and discrimination slopes (a_i) with respect to empirical success rates.
  - 2PL IRT latent deliberative ability (θ).
  - Mathematical boundary conditions: ensures Laplace smoothing and regularization to prevent division-by-zero or parameter divergence under 0% or 100% pass rates.
- `test_transcripts_pdf.py` (5 tests): Verifies programmatic generation of reviewable PDF transcripts via ReportLab, custom two-pass page numbering, and badge rendering.
- `test_pipeline.py` (7 tests): Validates end-to-end execution, CLI argument parsing, experiment directory generation (`data/expXXX`), config validation, report synthesis, and safety guardrails (aborting report generation if all ballots fail, or discarding single unparseable ballots with a `[!WARNING]` Data Quality Alert banner).

All tests can be executed from the repository root using:

`pytest` or `pytest -v`

A sample test execution run is shown below:

```
============================= test session starts ==============================
platform darwin -- Python 3.12.1, pytest-9.1.1, pluggy-1.6.0
rootdir: /path/to/simple-simulation-irt-evaluation-in-deliberation
configfile: pyproject.toml
testpaths: tests
```

```
tests/test_ballot_parser.py .......                                      [ 22%]
tests/test_deliberation.py .......                                       [ 45%]
tests/test_model_availability.py                                         [ 48%]
tests/test_pipeline.py .......                                           [ 70%]
tests/test_psychometrics.py ....                                         [ 83%]
tests/test_transcripts_pdf.py .....                                      [100%]

======================== 30 passed, 1 skipped in 3.86s =========================
```

5 Limitations and Open Questions

While this benchmark provides a standardized proof-of-concept for measuring artificial deliberation through psychometrics, nine structural limitations and open questions warrant explicit consideration:

1. The biggest assumption this experiment makes is that there are "ideal" policy outcomes to deliberative processes, and that the ones selected are valid candidates for this. Insofar as consensus is desired from deliberative democracy, this may not be detrimental or undesirable. But it is still a binarization of subjective deliberation that may not be robust or representative enough.

2. In the case of dyadic deliberative ability, the "success" condition for the deliberation is perfect consensus (both agents vote for the ideal choice) which is given a score of 1 and anything else 0. This may be too binary, with a possibility of awarding partial credit based on at-least 1 of the 2 agents voting ideally.

3. Only turn-based dyadic deliberations are simulated, with the possibility to extend to spontaneous/interjection-based deliberation involving more than 2 participants.

4. In a 4-alternative forced-choice instrument, uniform random guessing yields an expected accuracy of $P_{\text{guess}} = 0.25$ and an expected Distractor Choice Ratio of $\text{DCR}_{\text{chance}} = 1.000$. Consequently, empirical DCR values between 0.1 and 0.5 reflect modest secondary attraction alongside the dominant compromise rather than widespread electorate capture. The primary empirical divergence at baseline is the contrast between the Disaffected Populist and the other three personas: while the Technocrat, Ideologue, and Everyman selected Option A in 100% of trials (15/15 in `gpt-oss-120b` and 14/14 valid in `Muse-Spark`), the Populist selected Option A at only 46.7% (7/15) in `gpt-oss-120b` and 42.9% (6/14) in `Muse-Spark` (two-tailed Fisher's exact test $p = 0.002$ compared against any single 100% baseline persona such as 15/15 in `gpt-oss-120b` or 14/14 in `Muse-Spark`, or $p = 2.5 \times 10^{-6}$ against the three other personas pooled in `gpt-oss-120b`, and $p = 1.6 \times 10^{-6}$ in `Muse-Spark`). While this 43%–47% baseline compromise rate exceeds the 25% chance floor, the remaining distractor choices concentrated predominantly on Distractor B (Status Quo Inertia; 5 of 8 distractor choices in `gpt-oss-120b` and 5 of 8 in `Muse-Spark`). In models where baseline Populists selected Option A uniformly (such as `DeepSeek` and `Qwen3.5-9B` with reasoning disabled), the instrument registered ceiling saturation without distractor pull.

5. Across the benchmark battery, Option A (Integrative Compromise) is systematically longer (averaging 31.5 words versus 20.5–23.0 words for distractors, a +43% length differential) and consistently contains classical deliberative hedge phrasing (*"while preserving..."*, *"preventing surface acquisition while realigning..."*) alongside real-world institutional policy frameworks (COMPASS, MAS/SGX taxonomies). While Option A also represents the most substantively balanced policy resolution, relatively more competent models may identify Option A via syntactic complexity and length heuristics rather than genuine deliberative convergence. Nonetheless, the Disaffected Populist's divergence at baseline (selecting distractors 53% to 57% of the time under identical option wording) indicates that item surface features alone do not uniformly drive model selections. Future benchmark revisions should enforce strict character-count balancing and distractor syntactic parity.

6. Because each model dataset is calibrated independently with regularized zero-mean priors ($\theta \sim \mathcal{N}(0, 2), b \sim \mathcal{N}(0, 2)$), the resulting trait ($\theta$) and difficulty ($b$) estimates reside on separate, unlinked latent scales. Comparisons of parameter values across models must be interpreted **strictly in terms of relative rank orders and structural distributions**, rather than direct arithmetic subtractions. Full scale equating would require joint concurrent calibration across pooled responses.

7. To preserve clean dialogue transcripts and strict single-letter ballot formatting, reasoning effort was constrained to low or disabled across models. Future work should evaluate models under stronger reasoning regimes (such as medium or high reasoning budgets) paired with more robust postprocessing pipelines, either via auxiliary LLM reformatting passes or algorithmic rules, to test whether extended internal deliberation enhances persuasive quality without compromising format adherence.

8. Assigning "High Policy Knowledge" to the Technocrat versus "Low Policy Knowledge" to the Populist manipulates conversational register, rhetorical tone, and emotive posture, but does not alter the underlying model's parametric knowledge weights. Future work should evaluate whether incorporating retrieved local knowledge corpuses or parameter adapters induces deeper behavioral divergence.

9. While overall speaking initiative is balanced across the 8-dyad matrix (each persona initiates 2 dyads and responds in 2 dyads) and speaking-order reversals were explicitly tested for two key pairings ($\boldsymbol{D}_2/\boldsymbol{D}_6$ and $\boldsymbol{D}_4/\boldsymbol{D}_8$), turn order remained fixed for the remaining pairings ($\boldsymbol{D}_1, \boldsymbol{D}_3, \boldsymbol{D}_5, \boldsymbol{D}_7$). In our standardized runs, consensus saturated at ceiling

across nearly all dyadic pairings ($\boldsymbol{D}_2$ and $\boldsymbol{D}_6$ both achieved full or near-full consensus, as did $\boldsymbol{D}_4$ and $\boldsymbol{D}_8$). Consequently, turn-order effects remain largely untestable at ceiling due to insufficient outcome variance. Conversational initiative could subtly influence rhetorical framing in longer or more contentious deliberations where responses do not saturate at ceiling.

10. The benchmark architecture executes conversational turns and secret ballot voting via stateless API calls, where the citizen system prompt and cumulative conversational dialogue history are re-injected on every turn. The simulated agents lack an autonomous memory stream, reflection architecture, or persistent internal belief-state across deliberations (such as those formulated in `Park et al. 2023`). While this ensures exact session isolation and prevents cross-trial leakage, it abstracts away how real citizens accumulate, consolidate, and retrieve lived experiences over time. Future experiments should evaluate whether equipping personas with stateful memory architectures and reflective memory synthesis influences deliberative malleability and resistance to cognitive traps.

11. In the present benchmark design, both conversational participants in any given dyad are backed by the identical underlying LLM checkpoint. While this cleanly isolates architecture-level psychometric profiles, it introduces intra-model stylistic correlation, shared inductive biases, and mutual sycophancy. These issues have been discussed in `Veri 2026` and `Wang et al. 2025`.

12. Beyond the presence of conversational dialogue, the pre-deliberation baseline differs from post-deliberation voting across several structural dimensions: the baseline ballot contains no conversational transcript, includes an explicit null indicator ("No prior deliberation took place. This is a baseline pre-deliberation evaluation."), operates on $N_0 = 15$ ballots per persona rather than $N_{\text{post}} = 36$, and post-deliberation ballots within each dyad share a joint transcript context. A placebo or scrambled-transcript control would more cleanly decouple conversational exposure from prompt-length and context-framing effects.

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A Qualitative Deliberation Transcripts: Dyad D2 on PMET Insecurity

This appendix presents the multi-turn conversational transcripts for Dyad $\boldsymbol{D}_2$ (Technocratic Pragmatist P_1 vs. Disaffected Populist P_3) on `ITEM_02` (Foreign PMET Salary Thresholds and Local Career Progression) across Trial 1 (recorded as `"trial_index": 0` in the benchmark archives) of all four evaluated model architectures. These transcripts illustrate the qualitative dynamics, localized idioms, and format compliance variations discussed in Section 3.

A.1 DeepSeek-V4.1-Flash-1017 (exp000)

Turn 1 — Technocratic Pragmatist (P_1):
The salary threshold should be treated as a calibrated filter, not a symbolic barrier: set high enough to prevent wage undercutting in mid-career PMET roles, but differentiated by sector and reviewed against real vacancy, wage, and hiring data. I would pair any threshold adjustment with mandatory transparency on local applicant flows and a firm-level knowledge-transfer and succession-planning requirement for firms relying heavily on foreign PMETs. The common ground is a data-driven bargain: openness where genuine specialization is scarce, and measurable local progression where it is not. If we can agree on the metrics first, we can negotiate the threshold and complementary local-pipeline measures without trading away either competitiveness or citizen trust.

Turn 2 — Disaffected Populist (P_3):
Look, a "calibrated filter" sounds fine on paper, but ordinary mid-career workers don't get hired by a dashboard—they get overlooked while companies say no locals were available. I'll meet you halfway on transparency only if it's company-level, public enough to matter, and backed by real penalties when firms keep importing PMETs without promoting locals. Set the salary threshold high enough that it actually stops wage undercutting, and make "knowledge transfer" mean named local successors and actual progression, not another HR form. If a company can't prove it's building local careers, it shouldn't keep getting the easy route to foreign hires.

Turn 3 — Technocratic Pragmatist (P_1):
I can accept company-level public disclosure and real penalties, provided we define the metrics precisely: local applicant-to-interview-to-offer ratios, PMET wage bands, and named successors with time-bound progression milestones. On the threshold, I would anchor it to the upper quartile of comparable local PMET wages by sector, not a single national number, so it genuinely blocks undercutting without shutting out scarce specialists. Enforcement should escalate—warning, published remediation plan, then suspension or curtailment of foreign-hiring privileges for firms that repeatedly fail. That gives you visible accountability and gives firms a predictable rulebook, which is the pragmatic bargain I can defend.

Turn 4 — Disaffected Populist (P_3):
I can live with your upper-quartile-by-sector threshold if it's published in plain English and updated, not buried in some ministry appendix nobody reads. But company-level disclosure has to show real names and real promotions, because "local applicant ratios" can be gamed with phantom interviews while the same foreign hire stays for years. And your penalties need teeth from day one—published warning, then a hard freeze on new foreign PMET passes, not just another remediation plan that drags on. If you give me that, I'll back the bargain: openness for genuine scarce specialists, but no easy route for firms that won't build local careers.

A.2 openai/gpt-oss-120b (exp001)

Turn 1 — Technocratic Pragmatist (P_1):
The foreign PMET salary thresholds should be calibrated to attract the specific skill sets Singapore needs while ensuring that the differential does not erode the progression ladder for local PMETs. We can achieve this by tying the thresholds to measurable productivity gains and mandating a structured knowledge-transfer component for each foreign hire. A transparent, data-driven review every two years would allow us to adjust the thresholds in line with labour market outcomes. This approach balances openness with safeguards for local career development.

Turn 2 — Disaffected Populist (P_3):
Look, we can't keep letting foreign PMETs swoop in on jobs just because they fetch a higher salary on paper – my sister's husband was passed over for a senior role even though he's been with the company for years. If you want to keep the hub-status, set a clear cap: any foreign hire must earn at least 30% above the median local salary ***and*** *the company must pledge a mentorship plan that guarantees a local employee will be promoted within two years. Let the Ministry audit those promises and fine firms that cheat, so we keep the talent flow but protect our own ladder.*

Turn 3 — Technocratic Pragmatist (P_1):
I agree that a clear, enforceable framework is essential, and a salary differential tied to measurable productivity can serve that purpose while avoiding arbitrary caps. Let us set the foreign-PMET threshold at a calibrated premium—e.g., 20% above the median local salary for comparable roles—subject to a mandatory, auditable knowledge-transfer plan that includes a documented mentorship timeline and a measurable promotion target for a local employee within two years. The Ministry of Manpower can conduct biennial reviews, adjusting the premium based on actual productivity gains and promotion outcomes, and impose graduated fines for non-compliance. This structure preserves Singapore's hub appeal, ensures tangible skill transfer, and protects the career progression of our local PMETs.

Turn 4 — Disaffected Populist (P_3):
I get that you want numbers and reviews, but the truth is families are watching real promotions slip away while foreign hires sit on fat paychecks. Let's fix the premium at 30% above the local median – that's a real bite-off for companies – and make the mentorship promise concrete: a written development plan, quarterly check-ins, and a guaranteed promotion slot for a Singaporean within 18 months, otherwise the firm pays a steep fine that goes into a retraining fund. The Ministry can still do a biennial audit, but the penalty should be enough to make firms think twice before sidestepping locals. This way we keep the talent we need, but we also protect the ladder for our own people.

A.3 Qwen/Qwen3.5-9B (exp002)

When evaluated with reasoning disabled (`enable_reasoning: false`), `Qwen/Qwen3.5-9B` generated complete, unmediated dialogue across all 4 turns without leaking internal planning scratchpads or exhausting generation token limits:

Turn 1 — Technocratic Pragmatist (P_1):
Recognizing our shared goal of maintaining Singapore's competitiveness as a global hub while safeguarding our domestic talent pool, I propose we focus on refining the current framework rather than adopting rigid measures that disrupt market dynamics. Instead of setting arbitrary salary caps, we should strengthen the reliance on the Progressive Wage Model and mandatory localisation quotas for leadership roles to ensure that knowledge transfer is structurally guaranteed alongside foreign hiring. By aligning foreign talent acquisition with specific skill gaps identified through empirical data, we can mitigate wage suppression concerns without stifling the innovation required for our economy to thrive. Let us explore a mechanism where firms must demonstrate a clear pathway for local upskilling before accessing top-tier foreign talent, ensuring both fiscal prudence and fair career progression for our citizens.

Turn 2 — Disaffected Populist (P_3):
I hear you talking about "empirical data" and "market dynamics," but you know what really matters to us on the ground? It's whether a local junior manager actually gets to sit in the boardroom or if they're just being used as a temporary stopgap before a foreigner takes over the permanent seat. If companies say they will upskill locals but don't fire their top foreign hires when we're ready, that's just empty promises that make us feel like second-class citizens in our own city. Let's make the localisation quota a strict, non-negotiable rule with real penalties for firms that don't prove they've actually transferred skills before hiring more expensive expats, because our families can't wait for a perfect economic model to survive; they need guaranteed jobs today.

Turn 3 — Technocratic Pragmatist (P_1):
I understand your frustration that promises of future upskilling often feel like empty rhetoric when locals are immediately sidelined by foreign hires; however, imposing strict hiring bans without a proven track record of knowledge transfer risks crippling our small domestic talent pool and scaring away the very global firms that drive our economy. Rather than rigid caps that might stifle competitiveness, we should mandate a transparent, audited "localisation credit system" where firms earn foreign hiring quotas only after demonstrably mentoring locals into senior roles with measurable performance milestones. This approach ensures that every dollar spent on expat salaries directly funds the elevation of our citizens, turning your concern about empty promises into a verifiable operational requirement for all corporate entities. By tying executive compensation bonuses to these specific local progression metrics, we align the financial incentives of top management with the long-term career stability of our workforce, satisfying both the need for global expertise and the demand for fair local opportunity.

Turn 4 — Disaffected Populist (P_3):
You're right that tying bonuses to our actual promotion is the only way to stop management from treating our careers as an expense they can dump whenever a foreigner arrives. But until they prove that their "localisation credits" are real and not just paperwork filed to get a tax break, we won't accept that foreign hires will ever be forced out of those top jobs. We need a hard rule where a local must be officially certified as ready for the role before a single dollar is spent on bringing in a replacement, or else these companies face immediate fines and public shaming. Let's make this a no-questions-asked deadline so that when my colleague finally gets the promotion they've been waiting for, no one can claim the company wasn't ready to let them lead.

A.4 meta-models/Muse-Spark-Open-816B-A42B (exp003)

Turn 1 — Technocratic Pragmatist (P_1):
As a citizen who values both our economic competitiveness and social compact, I recognise that maintaining Singapore's status as a regional headquarters hub requires access to specialised global talent, but that access must not come at the cost of hollowing out our local PMET pipeline. The salary threshold for Employment Passes and S Passes is our most precise calibration tool - set too low it risks wage suppression, set too high it risks hollowing out investment. I believe we can find common ground by moving beyond a binary open-versus-closed debate and instead focusing on how to make the threshold more dynamic and accountable. I propose we explore a framework where thresholds are sector-calibrated and explicitly tied to verifiable local capability transfer.

Turn 2 — Disaffected Populist (P_3):
Look, with respect, that "precise calibration tool" talk sounds very neat in a boardroom but on the ground my friends in their 40s are getting lowballed or replaced and told to train their replacement. I'm sick of hearing we need "global talent" while locals can't even get an interview for jobs we are qualified for. If you want a sector-calibrated threshold then fine, but it has to be set HIGH and tied to real penalties — like for every EP you hire, you must prove you hired and promoted a Singaporean with a proper handover, not just wayang.

Turn 3 — Technocratic Pragmatist (P_1):
I hear your frustration, and you're right that a threshold means little without enforcement — wayang mentorship and low-balling mid-career locals undermines both fairness and our long-term talent pipeline. I can support a higher, sector-calibrated threshold — for example setting EPs at a clear premium above the local median for that specific sector and age band — but only if we pair it with a verifiable quid pro quo. For every EP, the firm should be required to file a structured capability-transfer plan with KPIs, local understudy promotion timelines, and be subject to MOM audit with escalating levies or quota curbs for non-compliance.

Turn 4 — Disaffected Populist (P_3):
Okay, that sounds better — premium above median for age band plus real KPIs and audit, not just self-declare. But I need to see teeth: publish the firms that fail, hit them with higher levies, and cut their EP quota next year, otherwise it's still wayang. And protect the 40s and 50s — if you hire an EP at say $12k, you better show a Singaporean in that same team getting trained to take that role in 2-3 years with pay bump, not stuck as forever number two. You agree to that kind of transparency and timeline, then I can live with your higher sector threshold instead of just shutting the door.